

# Errors in Claimed Complete Proofs (and Disproofs) of the Collatz Conjecture: A Catalog and Taxonomy of Twelve Cases

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## Abstract

The Collatz conjecture attracts an unusually large volume of claimed complete proofs and disproofs, mostly circulated through preprint servers and low-scrutiny venues rather than peer-reviewed mathematics journals. We report a systematic review of twelve such claims encountered during a broader literature survey, each read in full, independently reconstructed, and checked computationally wherever the claim admits a numerical test. None of the twelve is a valid proof or disproof. The contribution of this note is not the count itself but a taxonomy of the ways they fail: two categories account for ten of the twelve cases (circular reasoning that smuggles the conjecture’s conclusion into a definition or premise, three cases; illegitimate generalization from one hand-picked adversarial sequence to all sequences, two cases), and a single recurring fallacy — treating an aggregate, average, or asymptotic property as a deterministic, pointwise guarantee for one specific trajectory — accounts for five cases by itself, disguised in turn by Baker’s theorem, a random-walk heuristic, a 2-adic Lyapunov argument, Big-O notation, and static 2-adic density. The remaining two cases are an elementary error unrelated to the conjecture’s real difficulty, and a genuine rigor gap that, unlike its closest analogue in the catalog, produced no computational counterexample. Ten of twelve failures converge, by superficially unrelated routes, on the same structural obstruction that no known method resolves: the impossibility of bounding adversarial runs of increasing steps without, at some point, assuming what is to be proved.

## 1 Introduction

The Collatz conjecture has been open for close to ninety years, and its statement is simple enough that it draws an unusually large number of claimed complete solutions from outside the community of professional number theorists. Almost all of them are circulated through preprint repositories, personal websites, or journals with no established reputation in pure mathematics, rather than peer-reviewed venues with specialist referees. None has been accepted by the mathematical community. This is not, by itself, evidence that any particular claim is wrong — but it sets the correct prior: a claimed complete proof (or disproof) of a problem this resistant, from a venue without specialist peer review, should be read with the working assumption that it contains a genuine gap, most likely at exactly the point where the known “structural obstruction” to the conjecture would have to be overcome (the impossibility, with any method known today, of bounding how long an adversarial run of increasing steps can be — see [1] for the standard account of why this is hard).

This note catalogs twelve such claims, encountered over the course of a broader literature survey conducted alongside an unrelated research program on a  $qx + 1$  generalization of Tao’s Syracuse

measure [2]. Each claim was read in full, its central argument reconstructed independently, and checked computationally wherever it admits a numerical test; several of the more delicate judgment calls were also cross-checked against an independent large-language-model mathematical reviewer. None of the twelve survives scrutiny as a valid proof or disproof.

The point of this note is not the bare fact that twelve amateur claims are wrong — that would be unremarkable and uninteresting on its own. The point is *how* they fail. Section 3 groups the twelve cases into five categories by the shape of their central error, together with a brief summary of each case; three categories, covering ten cases, converge on the same structural obstruction from routes that look, on the surface, entirely unrelated to one another. Section 4 discusses what this convergence suggests about why the conjecture resists these particular styles of attack, and what the two exceptions to the pattern reveal by contrast. Section 5 concludes.

To be clear about what this note is *not*: it is not a claim about the authors’ competence or good faith, several of whom are explicit and careful about the limits of their own arguments elsewhere in the same text (correctly labeling an open step a “conjecture” rather than a “theorem”, for instance). The focus throughout is strictly the mathematical content of the specific error identified in each case.

## 2 Methodology

Each of the twelve claims was: (i) read in full, not just abstract and conclusion; (ii) reconstructed as an explicit chain of premises and inferences, so that the exact step carrying the logical weight of “proves” or “disproves” could be isolated; (iii) checked computationally whenever the claim, or the step suspected to be the gap, admits a numerical test — either by direct reimplementing of the claimed identity, or by constructing a targeted counterexample; and (iv), in the handful of cases where the mathematical judgment involved was subtle enough to warrant a second opinion, cross-checked against an independent large-language-model reviewer with no access to our own working conclusion in advance. Full technical detail for every case — the exact reconstructed argument, the computational check, and where applicable the source code — is kept in individual case files referenced in Section 3; this note summarizes rather than reproduces that material.

## 3 Taxonomy of recurring error patterns

An initial pass at this taxonomy, built only from each case’s title and a one-line status summary, proposed seven categories. It did not survive contact with the full texts: one case (Boyle, §3.3) fit none of the seven, and reading the remaining cases in full showed that it, together with four others, was in fact the same error dressed in unrelated formal apparatus. The taxonomy below reflects the full reading and has five categories, one of which — Section 3.3 — accounts for five of the twelve cases by itself.

### 3.1 Category A: circular reasoning

The argument’s central object, or its central step, is well-defined or valid only under the assumption that the conjecture already holds — so the “proof” does not establish termination, it merely restates it in different notation. Three cases:

**Getachew (2025)** [3] constructs a reverse tree of the Collatz map and defines a “parent” relation that turns out to be, term for term, identical to the forward Collatz map itself ( $\text{parent}(m) = m/2$  for even  $m$ ,  $3m+1$  for odd  $m$ ). The paper’s Lemma 4.3, “every path back to the root is finite,”

is then logically equivalent to the conjecture itself, not a consequence of the tree’s acyclicity and unique-parent structure as claimed: acyclicity and a unique parent guarantee that a path never repeats or branches, never that it is finite. The paper’s covering theorem (Theorem 5.1) fails for a parallel reason: it proves a universal arithmetic decomposition (odd number times a power of two) that we confirmed computationally never actually uses the branching rule.

**Spencer (2025a)** [4] builds a different reverse tree, indexed by residue classes modulo  $2 \cdot 3^J$ , with a ternary-counter argument partially formalized in Lean. It correctly proves that every primitive residue class is “occupied” at every finite scale  $J$  — but its Theorem 15.3 then concludes that every specific integer has a finite reverse address, a step never justified: “residue occupied” does not imply “exact value reached.” We confirmed this gap directly: the residue class of  $n = 27$  is occupied in the real tree built to depths 6 and 8, but only by representatives on the order of  $10^6$ – $10^8$ , never by 27 itself.

**Yun (2026)** [11] contains three independent circular arguments, any one of which alone invalidates the paper’s claim. Its central “rank function”  $r(x) := \min\{n \in \mathbb{N} : f^{(n)}(x) = 1\}$  is defined only for values of  $x$  whose orbit already reaches 1 — if some orbit never reached 1, exactly the possibility the conjecture needs to exclude,  $r(x)$  would simply not exist, so the well-founded descent argument built on it presupposes its conclusion. We confirmed this by applying the identical construction to  $g(x) = 2x$ , a map that provably diverges for every  $x > 0$ : the analogous rank function is undefined for all 18 tested values, exactly as it would be for  $f$  if some orbit failed to reach 1. Separately, the paper’s cycle-exclusion argument (Theorem 6.1) asserts that “the only known fixed point is  $x = 1$ ” as if this excluded other cycles, and its “informational compression” argument (§5, §9.2.3) is a non sequitur about cardinalities of infinite sets that is compatible with any long-run behavior, not specifically convergence to 1.

### 3.2 Category B: illegitimate generalization from a single sequence

The argument shows, correctly, that one specific hand-picked sequence of increasing and decreasing steps has a certain property, then extends the conclusion to all sequences without bounding how long an adversarial run of increasing steps could be — exactly the step the structural obstruction makes hard. Two cases:

**Santos (2018)** [6] introduces a binary “moves”/“strong reductions” notation equivalent to the ordinary 2-adic valuation sequence of a Collatz orbit. Its core argument (§2.6) picks a hypothetical  $K > 30$ , *assumes a specific fixed sequence* of steps (two increasing moves followed by three decreasing ones, with hand-chosen divisors), shows algebraically that  $K$  decreases along *that particular sequence*, and then generalizes this to every  $K > 30$  without ever showing that adversarial sequences of increasing steps cannot be made arbitrarily long.

**Halemane (2025), the CTUHSK theorem** [7] reformulates the standard reverse tree with new notation (Binary-Exponential-Ladders, operators  $D^\# / U^\#$ ) but adds nothing structurally new. Its necessary condition is a tautology (the set *defined* as reaching the trivial cycle does reach the trivial cycle); its sufficient condition, which carries the real weight of excluding extra cycles and divergent chains, has a decisive hole: the reductio argument (§10.2.1) assumes the predecessor of a minimal element of a hypothetical cycle is given by the paper’s formula with exponent  $v = 1$  specifically, when that formula in fact has infinitely many valid solutions, one per odd exponent  $v = 1, 3, 5, \dots$ ; only  $v = 1$  yields a smaller value (violating the assumed minimality), while every  $v \geq 3$  yields a larger one, generating no contradiction at all. We confirmed computationally that this is a general algebraic pattern, not an artifact of the paper’s chosen numerical example.

### 3.3 Category C: an aggregate or asymptotic property mistaken for a deterministic, pointwise guarantee

This category is the paper’s central empirical finding. Five superficially unrelated arguments — built respectively on Baker’s theorem, the classical random-walk heuristic, a 2-adic Lyapunov bound, Big-O asymptotic notation, and static 2-adic density — turn out to commit the identical error: a property that holds only as an average, an ensemble expectation, or an asymptotic bound is substituted, without justification, for a guarantee that must hold deterministically, at every point, along one specific trajectory. Three of the five papers below cross-reference this exact pattern in each other, independently of this catalog, which suggests it is a natural attractor for Collatz-proof attempts rather than an artifact of our classification.

**Mohammed [8]** builds a correct geometric density sieve and correctly excludes 1-cycles, then attempts to exclude nontrivial cycles via Baker’s theorem on linear forms in logarithms. The cycle equation genuinely forces  $M/P \rightarrow \log_2 3$  for a real, self-consistent cycle with large elements — but the paper substitutes  $M \approx 2P$ , the *ergodic, average* expectation for a generic odd integer (already known in this project as a baseline fact, not a cycle constraint), producing a check that constrains nothing.

**Boyle (2026) [9]** derives, via the standard random-walk heuristic, that the ensemble-average fraction of even steps along Collatz orbits is  $2/3$  — valid, at best, as an average over many random starting values  $n$  — and then substitutes this constant directly into an equation (Lemma 4.2) about one specific, fixed, hypothetical divergent trajectory, deriving a “contradiction” that depends entirely on this illegitimate substitution. We confirmed computationally that even among orbits known to converge, the fraction of even steps realized by any one individual trajectory varies substantially around  $2/3$  (mean 0.677, standard deviation 0.031 over 3000 sampled trajectories) — it is an aggregate statistic, not a constraint on any single orbit.

**Tynski (2026) [12]** builds an extensive and mostly correct formal apparatus (Clifford algebra, an explicit trace, established cycle-exclusion identities) around a “master theorem” unifying the Riemann hypothesis and the Collatz conjecture, but the exclusion of divergent orbits rests entirely on an axiom asserting a *deterministic* Lyapunov bound  $\log_2 n_m \leq \log_2 n_0 - m\delta + C\sqrt{m}$  for a universal constant  $C$ , justified circularly (the text derives the needed decay “once  $A_m/m > \log_2 3$  is forced,” the very property being argued for) and supported only by numerical verification up to  $n_0 \leq 3 \times 10^4$ . We tested the claim directly beyond that range: the minimal  $C$  required already exceeds the claimed universal bound of 5 for seeds  $10^4 \leq n_0 < 10^9$  (worst case 5.08), and known long-delay record-holders require  $C$  up to 8.34 — a direct computational refutation, not merely an absence of proof.

**The Syzdykov dispute (2025–2026) [13, 14, 15]** is a three-paper exchange rather than a single claim. The original note [13] asserts  $O(f(n)) \leq n$  for the Collatz function  $f$ , substitutes  $n = 2^m - 1$ , and declares  $O(f(2^m - 1)) \approx 3^m > 2^m - 1$  a contradiction refuting the conjecture. Lafontaine & Cheong [15] correctly identify the category error:  $O(\cdot)$  denotes a set of functions, not a scalar, so  $O(f(n)) \leq n$  is not well-formed; and even under a charitable pointwise reading, an asymptotic bound (“for some  $C, n_0$ , for all  $n \geq n_0$ ”) licenses no conclusion about the exact value of one specific  $n$  — the same confusion of an existential, aggregate bound with a pointwise, deterministic one seen in the other four cases in this category. Syzdykov’s reply [14] does not engage this point; it redefines  $O(f(n))$  after the fact to mean a scalar orbit length and reasserts the original conclusion as “already proved.”

**Barghout (2019) [16]** correctly proves, by elementary induction, that the 2-adic valuation of even integers (or of values produced by  $3n + 1$ ) follows a fixed, deterministic density pattern — a static fact about the *set* of even integers, not about any specific orbit. Its convergence argument

then imports this static density as if it were the *visit statistic* of one specific dynamical orbit, implicitly assuming the orbit equidistributes — precisely the unproved content of the conjecture, not a consequence of the density lemmas. We confirmed by direct simulation that individual orbits (e.g.,  $n = 9663$ ) can grow to roughly 2800 times their starting value before eventually decreasing, illustrating concretely why an “on-average” argument constrains no single trajectory.

### 3.4 Category D: an elementary error unrelated to the structural obstruction

**Roif (2026)** [10] constructs a three-point topological space  $S = \{0, 2, 1\}$ , reinterprets the integer 2 as an “empty container”  $\{\}$ , and proves a lemma asserting that any subset of the positive integers with asymptotic density zero must be empty. This is false as a general mathematical fact and is the only error in the catalog with no connection to the conjecture’s real difficulty: density zero does not imply finiteness, let alone emptiness. The set of perfect squares, or of powers of 2, has density zero and is plainly infinite. This single elementary error collapses the paper’s entire closing argument (its Theorem 9.1 and Corollary 9.2, which would otherwise complete the proof), independent of the remaining formal apparatus (Clifford-adjacent identities, a Lyapunov-style wave function), which is a correct but by-itself insufficient restatement of results already standard in the literature.

### 3.5 Category E: a genuine rigor gap without a computational counterexample

**Spencer (2026)** [5], by the same author as the Category A case in §3.1, uses a different reverse tree (infinite-degree, via the inverse of the accelerated map). Unlike its companion paper, building the real tree and testing coverage up to 10,000 found *no* gap in practice — apparent coverage failures at smaller computational budgets converged exactly to the root once checked without a magnitude cutoff. We nonetheless identified a genuine gap in the written argument: Theorem 14.1 proves occupation of residue *classes*, and Theorem 14.2 concludes occupation of individual *elements* without justifying the step — the same logical shape as the Category A case, but one that, so far, has not manifested as an actual failure under computational search. This distinction matters: not every gap in a Collatz proof attempt is equally fatal, even within the same author’s closely related arguments.

## 4 Discussion

Ten of the twelve cases — all of Categories A, B, and C — fail, by routes that look unrelated on the surface, at exactly the same point: none of them establishes a genuine bound on how long an adversarial run of increasing steps can be, the structural obstruction that no known method resolves. Category C alone accounts for five of the ten, and is the most striking part of the pattern: the same conceptual error — substituting an aggregate, average, or asymptotic property for a pointwise, deterministic guarantee — survives being dressed in five unrelated formal vocabularies (elementary number theory via Baker’s theorem, the classical random-walk heuristic, a 2-adic Lyapunov bound, Big-O asymptotic notation, and static 2-adic density). This is not an artifact of our own classification: three of the five source documents already cross-reference each other, and two of them cross-reference the random-walk heuristic explicitly, as instances of the same confusion, independently of this catalog.

The two cases outside this pattern are informative by contrast. Roif’s error (Category D) is an isolated, elementary mistake with no bearing on the conjecture’s actual difficulty — a reminder that not every error in a Collatz proof attempt is evidence about *why* the conjecture is hard, some are simply wrong for unrelated reasons. Spencer’s second paper (Category E) shows the opposite

lesson: an identical logical shape to a Category A failure (residue-class occupation conflated with individual-element occupation) can, in one instance, fail to manifest as an actual computational counterexample — underscoring that the taxonomy classifies argument *structure*, not severity, and that structurally similar gaps are not always equally fatal.

We note, finally, that the broader collection of Collatz-related papers surveyed alongside these twelve (outside the scope of this note) contains many papers that make no claim of a complete proof at all, and that are careful to label open steps as conjectural. The twelve cases here were selected precisely because they make the strong claim; the taxonomy above should not be read as suggesting that claimed-proof papers are typical of the wider literature on the conjecture.

## 5 Conclusion

All twelve reviewed claims of a complete proof or disproof of the Collatz conjecture fail under independent scrutiny, and none is close to valid. The value of this catalog is not the count but the concentration: half of the twelve cases, despite using entirely different mathematical vocabularies, commit the identical error of mistaking an aggregate or asymptotic property for a deterministic, pointwise guarantee, and three more commit one of two closely related errors (circularity, illegitimate generalization from a single sequence) that converge on the same known structural obstruction. This regularity is itself a small piece of evidence about why the conjecture resists elementary attack: the failure modes are not diverse, they repeat.

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